

AI for IoT - Capstone Project

Report Format & Guidelines (Experiment 8)

Group Structure

- Groups of **3-4 students**
 - Each group selects **one topic** from the Capstone Topic Bank (no repeats within a batch)
 - All members must be able to individually explain the full pipeline at viva - group work does not exempt individual understanding
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Report Format

- **A simple dashboard or visualization** — even a matplotlib plot comparing predicted vs. actual values, or a basic Streamlit app, to make the output feel like a real deliverable rather than just a notebook.
 - **GitHub repository submission** — alongside or instead of a zip file, to get students used to real-world workflows.
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Evaluation Suggestion

Component	Weight
Report quality & completeness	25%
Correct use of prediction model	20%
Correct use of anomaly/classification model	20%
Dashboard/visualization	10%
Demo / walkthrough	15%
Individual viva	10%

Section	Contents
Title Page	Project title, group members with roll numbers, course code, guide name
Abstract	150–200 words summarizing the problem, approach, and outcome
1. Introduction	Problem statement, motivation, why this domain needs an IoT-AI solution
2. Objectives	3–5 clear, measurable bullet points
3. Related Work / Existing Systems	2–3 real-world examples researched (builds on Practical 07 case-study skills)
4. System Architecture	Sensor layer → connectivity → processing → AI model → application layer, with a labeled diagram
5. Dataset / Data Simulation	How sensor data was simulated or sourced, and assumptions made
6. Methodology	Which AI technique(s) were used and why; preprocessing steps
7. Implementation	Key code snippets, tools/libraries used, screenshots
8. Results & Evaluation	Metrics (RMSE / accuracy / precision-recall / F1 depending on model), graphs, output screenshots
9. Discussion	What worked, what didn't, limitations of the simulation
10. Conclusion & Future Scope	Summary, and how the project could extend to real hardware
11. Individual Contribution	Table listing each member's specific contribution
References	Any papers, articles, or datasets referenced
Appendix	Full code listing

